

TOO DIFFUSE TO LOCALIZE: DIAGNOSING AND CORRECTING THE MSE BIAS IN VIDEO-GUIDED AMBISONIC SPATIALIZATION

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ABSTRACT

Video-guided first-order ambisonic (FOA) spatialization predicts the directional channels of a sound field from a mono track and its 360-degree video. We show that spatializers trained with the standard complex-spectrogram mean-squared error (MSE) are biased toward *diffuse* fields: the MSE-optimal prediction is a conditional mean whose directional energy is reduced by the conditional variance of the source direction given the video, and whose shape is a posterior average over plausible directions. On the spherical energy maps used by 360-degree benchmarks this lowers correlation (CC) and AUC even when the peak direction is right. A validation-selected gain on the directional channels recovers part of the gap, but we remove the bias at its source with a differentiable energy-map loss written in closed form from the STFT spatial covariance. On 6,553 YT-Ambigen test clips this raises CC from 0.738 to 0.781 and AUC from 0.899 to 0.920, beats the calibrated MSE model, and flattens the gain curve, while leaving the mono channel untouched. The benchmark’s camera direction, we further show, coincides with the target map’s peak: a zero-parameter plane wave placed there already scores 0.69–0.75, matching end-to-end generation. We therefore report all models with and without it.

Index Terms— Spatial audio, first-order ambisonics, video-to-audio, 360-degree video, loss-metric mismatch

1. INTRODUCTION

Video-to-audio generation now produces plausible, well-synchronized mono sound from silent video, from early autoregressive and diffusion models [1, 2, 3] to recent multimodal systems [4, 5, 6, 7]. For 360-degree video the sound must also *arrive from the visible direction*. First-order ambisonics (FOA) is the standard representation: an omnidirectional channel W and three directional channels (Y, Z, X) encode the field on the viewing sphere [8]. Recent systems generate FOA directly from 360-degree video [9, 10, 11, 12, 13], and benchmarks such as YT-Ambigen [10] score the result by decoding the predicted field to a spherical energy map and comparing it with the ground-truth map using saliency metrics (CC and AUC-Judd [14]).

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This paper isolates the *spatialization* half of the problem: given the mono channel W and the video, predict (Y, Z, X) . This is the setting of Morgado et al. [9] and the second stage of two-stage FOA pipelines. Our starting observation, first made on the spatialization stage of our two-stage system FoleyImmersive [15] and reproduced here on a compact spatializer trained with the usual complex-spectrogram MSE, is that the decoded energy map is far too *diffuse*: multiplying the predicted directional channels by a constant gain selected on validation data improves CC and AUC-Judd on the test split without touching W . A post-hoc gain is a symptom, not a method. We explain the symptom and remove its cause:

- **Diagnosis** (Sec. 3). MSE regression of zero-mean directional channels returns the conditional mean, whose energy is reduced by the conditional variance of the channels given the input and whose shape is a posterior average over directions. Both effects flatten the decoded map. We measure the shrinkage factor on validation data and show that the empirically selected gain exceeds its reciprocal, i.e., calibration compensates for smearing as much as for lost energy.
- **Calibration** (Sec. 4.1). A validation-selected diagonal gain on (Y, Z, X) is a cheap, auditable fix that can be attached to any FOA generator, but it can only restore contrast, not placement.
- **Metric-aligned training** (Sec. 4.2). Because the benchmark map is a quadratic form in the field, it can be computed from the 4×4 STFT spatial covariance in closed form. We add the resulting differentiable map correlation to the objective; it outperforms the calibrated MSE model and makes calibration nearly irrelevant.
- **Protocol** (Sec. 2). We show that the camera direction supplied by YT-Ambigen coincides with the peak of the ground-truth energy map, so that a zero-parameter plane wave placed at it scores CC 0.69–0.75, in the range of published end-to-end generation systems; we report all models with and without this input.

Related work. Visually guided upmixing of mono audio to spatial formats was introduced for 360-degree video and FOA by Morgado et al. [9] and for binaural audio in [16, 17], all with spectrogram regression losses. End-to-end generation of spatial audio from video followed with ViSAGE [10], OmniAudio [11], diffusion-based and streaming FOA synthesis [12, 13] and stereo or binaural variants [18, 19]; the energy-map metrics we target follow [10, 14]. In the parametric view of ambisonics [20, 8] a field is a direction

Table 1. Retrievable subset of the official video-disjoint YT-Ambigen split used in all experiments.

Split	Clips (of official)	Videos	Role
Train	48,512 / 81,594	2,093	model fitting
Validation	6,122 / 9,604	250	gains, checkpoints
Test	6,553 / 11,166	264	final metrics only

plus a diffuseness, and our diagnosis says that MSE training overestimates the latter. That MSE regression blurs multimodal targets is well known for images and video [21]; we show the audio-spatial counterpart, quantify it against a public metric, and fix it with a loss written on that metric.

2. TASK, BENCHMARK, AND METRICS

FOA and decoding. We use ACN order (W, Y, Z, X) with SN3D normalization. Following the YT-Ambigen evaluator [10], a field $A = (W, Y, Z, X)$ is decoded on a 2° spherical grid $\mathcal{U}$ (16,200 directions) by the real first-order harmonics $\mathbf{h}(\mathbf{u}) \in \mathbb{R}^4$, and the energy map is the RMS of the decoded signal over a time window,

$$E_A(\mathbf{u}) = \left(\frac{1}{T} \sum_t (\mathbf{h}(\mathbf{u})^\top A(t))^2 \right)^{1/2} = (\mathbf{h}(\mathbf{u})^\top Q_A \mathbf{h}(\mathbf{u}))^{1/2}, \quad (1)$$

where $Q_A = \frac{1}{T} \sum_t A(t)A(t)^\top$ is the 4×4 spatial covariance of the window. The second form is exact and is what makes the map cheap to differentiate (Sec. 4.2). Maps are computed over the whole clip, at 1 fps and at 5 fps, resampled with the benchmark’s equal-area rule, and scored against the ground-truth map by Pearson correlation (CC) and by the saliency AUC [14] (the Judd variant of the released code). Both metrics ignore the overall level of a map but are sensitive to its *contrast*, i.e., to the balance between W and the directional channels. We re-implemented the released evaluator exactly (CC = 1.000 on ground truth) and use it throughout.

Data. YT-Ambigen [10] contains 102K five-second 360-degree YouTube clips with FOA audio. We use the roughly 61,000 clips that were still retrievable in September 2026, keep the official video-disjoint split, and evaluate on all 6,553 retrievable test clips (Table 1). Audio is resampled to 16 kHz.

Camera direction. For every clip the benchmark provides the camera direction (elevation, azimuth) of the perspective view that the video model sees, and ViSAGE conditions on it. This input is not innocent: by construction, YT-Ambigen crops the perspective view around the arg-max of the audio energy map [10], so the camera direction is a function of the target audio. Converting it to a unit vector, we find that it lies at a median of 3.2° from the peak of the ground-truth whole-clip map, within 30° for 89% of the test clips, and a single plane wave placed at it already reaches CC 0.689 (Table 2). Two remarks follow. First, the sign of the Y axis in the benchmark’s released mono-to-FOA encoding script is opposite to the one its evaluator expects: with that convention the same plane wave scores only 0.291, which matches the published “Ambi Enc.” baseline (0.349 [10]); we choose the sign on training clips and use the corrected convention throughout. Second, learned models that receive the camera direction must be read against the plane-wave row. We therefore report every model both with the camera direction, as in the benchmark protocol, and without it, where localization must come from the video and the audio alone. No other quantity derived from the target audio is used at inference.

Visual input. Following [10], equirectangular frames are projected to a 224×224 perspective view at the camera direction and encoded by CLIP ViT-B/32 [22] at 4 fps, giving a 20×512 feature sequence per clip.

3. THE MSE BIAS: SHRINKAGE AND SMEARING

Spatializer. Let $\mathbf{F}$ denote the conditioning input (STFT of W , CLIP sequence, camera direction). A two-dimensional U-Net G_ϕ (8.4M parameters; FFT 1024, hop 256; FiLM conditioning of every block by the CLIP sequence interpolated to the STFT frame rate and by the direction vector) predicts complex ratio masks $M_c(t, f)$ such that the directional spectrograms are $\hat{D}_c = M_c \odot W$, $c \in \{Y, Z, X\}$. W bypasses the network, so the mono content is never modified. The standard objective is

$$\mathcal{L}_{\text{MSE}} = \mathbb{E} \|\hat{D} - D\|^2 / \mathbb{E} \|D\|^2, \quad (2)$$

with D the target directional spectrograms.

The conditional mean is diffuse. The minimizer of (2) is $\hat{D}^* = \mathbb{E}[D | \mathbf{F}]$. By the law of total variance, at every time–frequency bin

$$\mathbb{E} |\hat{D}^*|^2 = \mathbb{E} |D|^2 - \mathbb{E} [\text{Var}(D | \mathbf{F})] \leq \mathbb{E} |D|^2. \quad (3)$$

The directional channels are zero-mean and change sign with the source direction, whereas the video fixes that direction only up to the uncertainty of the visual localizer, of onset timing, and of off-screen or multiple sources. Every such ambiguity is absorbed by the conditional variance and removed from the prediction’s energy. Moreover, the *shape* of $\hat{D}^*$ is the posterior average over plausible directions, i.e., a smeared field rather than a scaled copy of the true one. We summarize the first effect by the shrinkage factor

$$\alpha = (\mathbb{E} |\hat{D}|^2 / \mathbb{E} |D|^2)^{1/2} \in (0, 1], \quad (4)$$

estimated per axis on validation clips, and the second by the least-squares gain $g^* = \mathbb{E}[\hat{D} \cdot D] / \mathbb{E} |\hat{D}|^2$, which equals $1/\alpha$ only if the predicted field is a scaled copy of the target.

Effect on the map. For a plane wave from $\mathbf{u}_0$ with $\hat{D} = \alpha D$, (1) gives $E_{\hat{A}}(\mathbf{u}) \propto |1 + \alpha \mathbf{u} \cdot \mathbf{u}_0|$ instead of $|1 + \mathbf{u} \cdot \mathbf{u}_0|$: the front-to-back ratio falls from ∞ to $(1 + \alpha)/(1 - \alpha)$. Because CC and AUC compare map shapes after normalization, a uniformly shrunk but otherwise correct field still scores well: scaling the ground-truth directional channels by 0.36 lowers test CC only from 1.000 to 0.953, and even $g = 0.01$ retains 0.768 (Table 2). The large deficits of learned spatializers therefore come from shrinkage *and* smearing, which a post-hoc gain cannot separate; conversely, a gain applied to a correct field hurts ($g = 2.75$: 0.902), because it over-sharpens the map and creates negative lobes. Table 2 also quantifies the prior: the average training-set map, used as a constant prediction, reaches 0.374 and a random direction 0.007, whereas a plane wave at the benchmark camera direction reaches 0.689 (0.746 when over-sharpened by $g = 2$). Learned models that receive the camera direction must be read against this row, and a wrong-video control (Sec. 5.3) separates genuine localization from the prior.

4. TWO CORRECTIONS

4.1. Post-hoc spherical calibration

The cheapest correction leaves the trained model untouched and rescales the directional channels before decoding,

$$(Y', Z', X') = (g_y Y, g_z Z, g_x X), \quad W' = W, \quad (5)$$

where the gains maximize $J = \text{CC}_{\text{all}} + \text{CC}_{\text{1fps}}$ on validation predictions (a shared-gain grid followed by per-axis coordinate descent). No test clip is used for selection. Calibration is attractive as a diagnostic and as a drop-in fix for any FOA generator, but it is global: one gain per axis for all clips, frames and frequencies, whereas the conditional variance in (3) varies with the scene, and no gain can un-smear a posterior average.

4.2. Metric-aligned training

The mismatch originates in (2), so we correct the objective. By Parseval, the covariance in (1) can be accumulated in the STFT domain, $Q_A \propto \sum_{t \in \text{win}, f} \Re\{A(t, f)A(t, f)^H\}$, so the full 2° map of a window costs one 4×4 accumulation and a $16,200 \times 4$ quadratic form. We evaluate (1) for the 1 fps and 5 fps windows of every training clip, for the prediction $\hat{A} = (W, \hat{D})$ and the target A , and minimize

$$\mathcal{L}_{\text{map}} = 1 - \overline{\text{CC}_w}(E_{\hat{A}}, E_A), \quad (6)$$

the mean over windows of one minus the Pearson correlation of the two maps, weighted by $\cos(\text{latitude})$ as a smooth surrogate of the benchmark’s equal-area resampling. The total objective is $\mathcal{L} = \mathcal{L}_{\text{MSE}} + \lambda \mathcal{L}_{\text{map}}$: the MSE term keeps the fine spectral structure and the phase of the directional channels, while the map term compares normalized map *shapes*. Under uncertainty the MSE is minimized by shrinking, because any committed directional energy raises the expected error, whereas (6) is minimized by a field whose decoded contrast and placement match the target’s; the conditional mean, being flatter than the target, is penalized, so the two terms pull in different directions exactly where the ambiguity is largest. As ablations we also train with the map loss alone and with an *energy-ratio* penalty $\lambda_e (\log \mathbb{E}|\hat{D}_c|^2 / \mathbb{E}|D_c|^2)^2$ that only restores the scale, i.e., what calibration can do during training.

5. EXPERIMENTS

Setup. All rows share the split of Table 1, the (W, Y, Z, X) convention, the decoder (1) and the evaluator of [10]. The mono input is the ground-truth W , which isolates spatialization from sound generation. Every model uses the same U-Net, AdamW (learning rate 2×10^{-4} , weight decay 10^{-2}), batch 16, bf16 autocast and a budget of 7 h on one RTX 4090 ($\approx 80\text{K}$ steps; the no-video model 5.5 h, the two models without the camera direction 3 h). The checkpoint with the best validation J is kept; validation uses a fixed 600-clip subset during training and all 6,122 clips for gain selection and the shrinkage statistics. 95% bootstrap intervals are computed over test clip identifiers (2,000 resamples).

5.1. Main results

Table 3 gives the test results. With the camera direction, the MSE spatializer reaches CC 0.738, i.e., it barely improves on the plane wave placed at that direction (0.689, or 0.746 after over-sharpening), and its validation-selected gain adds +0.023 CC and +0.011 AUC (bootstrap 95% intervals of the paired difference: [0.021, 0.025] and [0.010, 0.012]), the calibration effect that motivated this work. The energy-ratio penalty raises the shrinkage factor from 0.60 to 0.73 but leaves CC unchanged (0.743): restoring energy without restoring placement does nothing for a shape metric. The map loss does what neither gain nor energy matching can: $\lambda = 3$ gives 0.775 and the map loss alone 0.781, above the *calibrated* MSE model on every metric without any calibration (paired bootstrap of the per-clip difference in CC: +0.013 [0.007, 0.019] and +0.020 [0.015, 0.024]; in

Table 2. Non-learned reference rows on the 6,553-clip test split (ground-truth W). g scales the directional channels; “cam.” is the benchmark camera direction.

Field	CC_{all}	AUC_{all}	CC_{1fps}
Ground truth ($g = 1$)	1.000	1.000	1.000
Ground truth, $g = 0.36$	0.953	0.988	0.953
Ground truth, $g = 0.1$	0.860	0.953	0.862
Ground truth, $g = 0.01$	0.768	0.911	0.771
Ground truth, $g = 2$	0.953	0.994	0.953
Ground truth, $g = 2.75$	0.902	0.988	0.901
Training-set mean map (no input)	0.374	0.717	0.359
Plane wave, mean training direction	0.237	0.620	0.224
Plane wave at cam., released convention	0.291	0.649	0.280
Plane wave at cam., corrected, $g = 1$	0.689	0.882	0.657
Plane wave at cam., corrected, $g = 2$	0.746	0.919	0.712
Plane wave, random direction	0.007	0.504	0.006

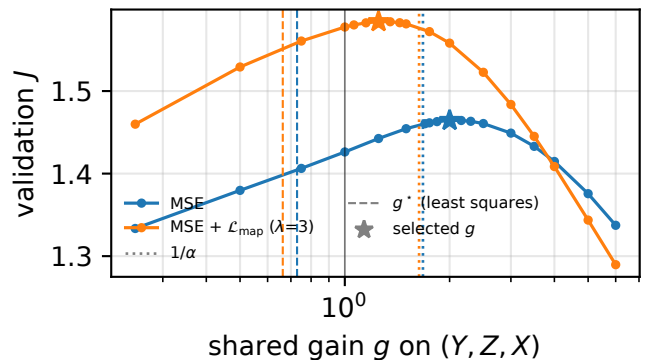


Fig. 1. Validation objective $J = \text{CC}_{\text{all}} + \text{CC}_{\text{1fps}}$ as a function of a shared gain on the predicted directional channels. Dotted: $1/\alpha$; dashed: least-squares gain g^* . The MSE model peaks far from $g = 1$; the map-loss model peaks near $g = 1$.

AUC: +0.006 [0.003, 0.009] and +0.010 [0.008, 0.012]). We use $\lambda = 3$ as the default ($\lambda = 1$ is dominated by the MSE term). Calibrating the map-loss models adds at most +0.006 CC, and their gain curves are flat (Fig. 1). Removing the video features from the map-loss model *improves* the test score to 0.828 (+0.054 [0.049, 0.059]; validation scores are equal), which we read as the CLIP sequence over-fitting to the 2,093 training videos while the camera direction transfers: with this input the visual content contributes nothing that the metric can measure. Every model that receives the camera direction outscores the published ViSAGE numbers, but the tasks differ—we are given W rather than generating it—so we make no generation claim.

Without the camera direction the picture inverts. MSE regression collapses to the prior (0.364, below the mean-map baseline 0.374; a gain adds only +0.013). The map loss lifts this to 0.500 (+0.123 [0.111, 0.134] over the calibrated MSE model, with a third of its training time), and its wrong-video control (0.14) confirms genuine per-clip localization. Dropping the video as well leaves 0.409: here the visual stream contributes +0.091 [0.079, 0.103] CC, whereas with the camera direction it was redundant. Video-driven FOA localization is therefore real but well below what the leaked direction alone supplies (0.69), which we read as the headroom this

Table 3. Learned spatializers on 6,553 test clips (ground-truth W as mono input). Gains g are selected on validation only; α is the validation shrinkage factor (4), g^* the least-squares gain. Point estimates; 95% bootstrap intervals are reported in the text.

Training loss	Correction	α	g^*	g	CC _{all}	CC _{1fps}	CC _{5fps}	AUC _{all}	AUC _{1fps}
<i>Reported by ViSAGE [10] (generation from silent video; different task, full test split)</i>									
SpecVQGAN + learned spatializer	–	–	–	–	0.619	0.587	0.547	0.848	0.828
ViSAGE, directional guidance	–	–	–	–	0.721	0.671	0.624	0.890	0.864
ViSAGE, directional + visual guidance	–	–	–	–	0.635	0.584	0.531	0.846	0.819
<i>This work — camera direction given (ground-truth W)</i>									
MSE	–	0.60	0.73	1	0.738	0.704	0.663	0.899	0.878
MSE	shared gain			2.00	0.762	0.726	0.685	0.910	0.889
MSE	axis gain			(2.0,1.7,2.0)	0.761	0.726	0.684	0.910	0.889
MSE + energy ratio	–	0.73	0.59	1	0.743	0.705	0.661	0.896	0.874
MSE + $\mathcal{L}_{\text{map}}$ ($\lambda=1$)	–	0.56	0.79	1	0.751	0.712	0.667	0.902	0.882
MSE + $\mathcal{L}_{\text{map}}$ ($\lambda=3$)	–	0.61	0.66	1	0.775	0.736	0.692	0.916	0.896
MSE + $\mathcal{L}_{\text{map}}$ ($\lambda=3$)	axis gain			(1.44,1.25,1.25)	0.781	0.743	0.697	0.920	0.900
$\mathcal{L}_{\text{map}}$ only	–	0.77	0.27	1	0.781	0.745	0.703	0.920	0.900
$\mathcal{L}_{\text{map}}$ only	axis gain			(1.82,1.59,1.38)	0.786	0.750	0.707	0.928	0.908
MSE + $\mathcal{L}_{\text{map}}$ ($\lambda=3$), no video	–	0.50	0.73	1	0.828	0.791	0.747	0.932	0.911
<i>This work — camera direction withheld (video + audio only)</i>									
MSE	–	0.72	0.34	1	0.364	0.349	0.331	0.682	0.674
MSE	axis gain			(0.96,1.89,2.17)	0.377	0.362	0.344	0.692	0.685
MSE + $\mathcal{L}_{\text{map}}$ ($\lambda=3$)	–	0.47	0.61	1	0.500	0.482	0.463	0.752	0.743
MSE + $\mathcal{L}_{\text{map}}$ ($\lambda=3$), audio only	–	0.10	0.25	1	0.409	0.391	0.371	0.711	0.703

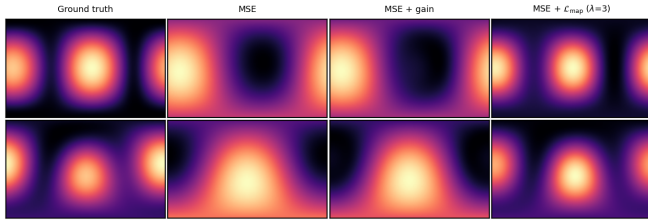


Fig. 2. Whole-clip energy maps (2° equirectangular grid) for two test clips chosen where the map-loss model gains most over MSE: ground truth, MSE, MSE with the selected axis gain, MSE + $\mathcal{L}_{\text{map}}$. The gain sharpens the MSE map but cannot move its mass.

benchmark still hides.

5.2. Shrinkage versus selected gain

Fig. 1 plots the validation objective against a shared gain. For the MSE model the selected gain, 2.0, is larger than $1/\alpha = 1.68$ and much larger than the least-squares gain $g^* = 0.73$: the field is not a scaled copy of the target, and the search over-sharpens it beyond the energy deficit to compensate for smearing. For the map-loss model α is similar (0.61), so the loss does not restore energy either; it restores the shape, after which the curve is flat between $g = 1$ and 2 and the selected gain is 1.25. This is the expected behaviour of a metric that ignores the level of a map: the target is placement, not power. Fig. 2 shows the same on two test clips chosen where the two models disagree most: the MSE map is smeared and its mass is misplaced, the gain sharpens it in place, and the map-loss model recovers both contrast and placement.

5.3. Sanity controls

Table 2 shows that the selected gain applied to the *ground-truth* field lowers CC ($g = 2$: 0.953), and Table 4 that predictions paired with

Table 4. Wrong-video control on the test split (camera direction given).

Model	Pairing	CC _{all}	AUC _{all}
MSE	matched video	0.738	0.899
MSE	wrong video	0.093	0.549
MSE	wrong video, axis gain	0.111	0.560
MSE + $\mathcal{L}_{\text{map}}$, $\lambda = 3$	matched video	0.775	0.916
MSE + $\mathcal{L}_{\text{map}}$, $\lambda = 3$	wrong video	0.140	0.582
MSE + $\mathcal{L}_{\text{map}}$, $\lambda = 3$	wrong video, axis gain	0.147	0.587

the *wrong* video (the field predicted for a clip from a different source video, scored against this clip’s map) fall to CC 0.09–0.14, far below the mean-map prior, and that the gain changes them by at most +0.02. The learned models therefore localize the specific clip rather than reproduce the prior, and the gain neither improves an already correct field nor rescues a mismatched one; its effect is specific to smeared, under-contrasted directional channels.

6. CONCLUSION

MSE-trained ambisonic spatializers are biased toward diffuse fields because the conditional-mean estimator discards the energy of every directional ambiguity and averages over the plausible directions. On YT-Ambigen the bias is measurable, a validation-selected gain recovers the contrast, and a differentiable energy-map loss derived from the STFT spatial covariance removes the bias during training, with and without the benchmark’s camera direction. We also showed that this camera direction is the peak of the target energy map and that the released encoder flips one axis, two facts that future comparisons on YT-Ambigen should account for. The analysis applies to any regressor of zero-mean spatial channels, including binaural upmixing [16, 17]; a global first-order field and a single frame-level map remain its limitations.

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